36021C01 – DATA MINING

Category: CC L-T-P-C: 3-0-0-3

Preamble

Data warehousing and data mining is one of the advanced fields of computer science involving Mathematics, Statistics, IT and Information Sciences in discovering new information and knowledge from large databases.

Course Outcomes

- CO1: Understand the basics of data warehousing and mining
- CO2: Learn data preprocessing, language, architectures, and concept description
- CO3: Learn association rules and related algorithms
- CO4: Learn classification and clustering rules and algorithms
- CO5: Know the latest trends in data warehousing and mining

Syllabus

UNIT I – INTRODUCTION AND DATA WAREHOUSING
Introduction, Data Warehouse, Multidimensional Data Model, Data Warehouse Architecture, Implementation, Further Development, Data Warehousing to Data Mining.

UNIT II – DATA PREPROCESSING, LANGUAGE, ARCHITECTURES, CONCEPT DESCRIPTION
Why Preprocessing, Cleaning, Integration, Transformation, Reduction, Discretization, Concept Hierarchy Generation, Data Mining Primitives, Query Language, Graphical User Interfaces, Architectures, Concept Description, Data Generalization, Characterizations, Class Comparisons, Descriptive Statistical Measures.

UNIT III – ASSOCIATION RULES
Association Rule Mining, Single-Dimensional Boolean Association Rules from Transactional Databases, Multi-Level Association Rules from Transaction Databases.

UNIT IV – CLASSIFICATION AND CLUSTERING
Classification and Prediction, Issues, Decision Tree Induction, Bayesian Classification, Association Rule Based, Other Classification Methods, Prediction, Classifier Accuracy, Cluster Analysis, Types of data, Categorization of methods, Partitioning methods, Outlier Analysis.

UNIT V – RECENT TRENDS
Multidimensional Analysis and Descriptive Mining of Complex Data Objects, Spatial Databases, Multimedia Databases, Time Series and Sequence Data, Text Databases, World

Wide Web, Applications and Trends in Data Mining.

Text Book

J. Han, M. Kamber, Data Mining: Concepts and Techniques, Morgan Kaufmann, 2001.

References

- Margaret H. Dunham - Data Mining: Introductory and Advanced Topics
- W.H. Inmon - Building the Data Warehouse
- Paulraj Ponniah - Data Warehousing Fundamentals